

“WHAT WE DO IS MAKE A SCAFFOLDING ON WHICH THE CELLS GROW.”

John Burke, American trauma surgeon, in *New York Times*, 1981

Artificial human skin, shown here, can be created by culturing human skin cells and growing them around a collagen matrix.

IN APRIL, NASA'S REUSABLE SPACECRAFT, the Space Shuttle, made its first full, orbital test flight. It was the first of 28 flights made by Space Shuttle *Columbia*, which tragically broke up on reentry in 2003. The Shuttle's main engines lifted the craft into orbit, with extra fuel provided by an external fuel tank; solid-fuel booster rockets provided extra thrust. Once empty, the boosters and the fuel tank detached from the Shuttle. The boosters were recovered and reused in future missions. NASA used **five space shuttles**, which made a total of 135 successful missions, lifting into orbit satellites as well as components for the **International Space Station**.

American company Osborne Computer Corporation released the **first successful portable computer**, the Osborne 1. This was followed by Japanese company Epson's **first laptop computer**, the HX-20, which weighed just 3.5lb (1.6kg) and worked on rechargeable

Preparing for launch

NASA's Space Shuttle *Columbia* sits on the launch pad at Kennedy Space Center in Florida, US. It is attached to the huge external liquid fuel tank and two slimmer solid-fuel boosters.



IBM-PC

One configuration of the IBM 5150 had two disk drives. There was no mouse, because the only interaction with the operating system was through typed commands.

referred to as the IBM-PC. The computer's operating system was PC-DOS, a version of MS-DOS (Microsoft Disk Operating System). Microsoft's enormously successful Windows

batteries. Xerox launched the 8010 Information System ("Star"), a workstation computer significant as the first commercially available computer with files displayed graphically as icons inside "window" folders, and an on-screen pointer controlled by a mouse. The **most important development in computer technology** this year was the release of the IBM 5150, usually

software (see 1985) was built around MS-DOS. The IBM-PC had a huge influence on the **development of the personal computer**. Its success led other companies to make clones—called IBM compatibles, which have dominated the personal computer market ever since. Companies could make IBM compatibles using off-the-shelf hardware, but the operating

system had to be licensed from Microsoft—making that company extremely successful.

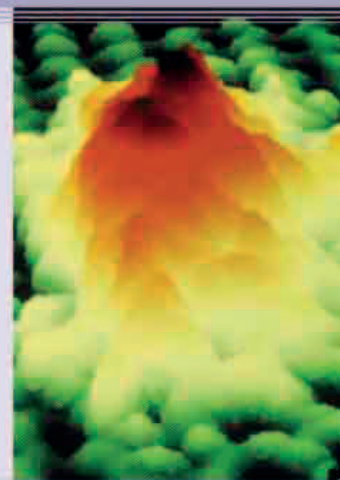
In April, Swiss physicist **Heinrich Rohrer** and German physicist **Gerd Binnig** succeeded in building the first **scanning tunneling microscope** (STM, see panel below)—an instrument with which scientists can produce accurate images of the atoms that make up solid surfaces. Also in April, American trauma surgeon **John Burke** and Greek-born American chemical engineer **Ioannis Yanas** developed the **first successful artificial skin** for patients with severe burns. Their skin was made with collagen from sharks and cows, sealed with a layer of silicone rubber. The collagen

formed a scaffold onto which the body could generate its own collagen, and build up new skin—and the silicone layer could then be removed.

Two separate teams, one in the UK and one in the US, developed the technology to isolate and culture **embryonic stem (ES) cells** from mouse embryos. ES cells are pluripotent—they have the potential to develop into any kind of cell. They can also replicate indefinitely. Human ES cells—first cultured in 1998—hold **great promise for future medical treatments**. For example, ES cells could be used to generate tissues for transplant, or implanted into the body, could help repair damage done by disease, injury, or aging.

SCANNING TUNNELING MICROSCOPE

At the heart of a scanning tunneling microscope (STM) is an extremely sharp metal tip that scans across a surface at very close range, measuring the electric current created by electrons "tunneling" across the gap between the surface and the tip of the probe. The color-enhanced STM image to the right shows a clump of gold atoms (yellow and brown) on a graphite surface (carbon atoms, green).



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THE MAXIMUM
**NUMBER OF
KILOBYTES OF
RAM MEMORY
IN THE IBM-PC**

August Microsoft releases first version of **MS-DOS** operating system, a branded version called PC-DOS given away with the IBM-PC

August 26 Space probe **Voyager 2** makes closest approach to Saturn and sends back stunning images

August 12 IBM launches the first successful personal computer, the IBM-PC

November Launch of the Epson HX-20, the world's first lightweight laptop computer

December 4 Vaccine for hepatitis-B virus approved in the US

December American molecular biologist Gail Martin reports that she had cultured pluripotent cells from mouse embryos, and she coins the term **embryonic stem cells**